

Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

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Abstract: Ecological networks provide a critical framework for understanding the architecture of biodiversity and predicting ecosystem responses to environmental change. However, the application of network ecology is often hindered by a lack of clarity regarding the assumptions inherent in different network representations. Here, we present a hierarchical framework that distinguishes between ‘metawebs’ (representing the fundamental feasibility of interactions) and ‘realised webs’ (representing interactions expressed in specific spatiotemporal contexts). We contrast our conceptual approach with recent data-centric reviews, focusing instead on the theoretical gradients that govern network construction. We identify five core processes that drive the transition from potential to realised interactions: evolutionary compatibility and co-occurrence, which define the feasibility of links; and abundance, diet choice, and non-trophic interactions, which determine their realisation. Furthermore, we map these processes onto a methodological spectrum of network construction, distinguishing between inductive approaches (*e.g.*, trait-matching and stochastic models) that infer structure from observation, and deductive approaches (*e.g.*, neutral and optimal foraging models) that generate structure from mechanistic first principles. By making explicit the assumptions and scale-dependent processes underpinning these different representations, this framework clarifies the scope of inference possible with each approach, ultimately facilitating more robust predictions of biodiversity dynamics in the anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

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1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.*, foundational studies on food web structure and robustness: [5]; [6]; [7]; [8]). Documenting interactions between and among taxa is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction and platform for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5]; [10]; [11]; recent discussions: [12]; [13]). Against

15 this, it is important to evaluate the value and the limitations of the various network
16 conceptualisations and how these relate to biodiversity concepts, such as community
17 structure or ecosystem function [14]. In this perspective we aim to provide an overview
18 of different **food web** representations, particularly how each representation embeds
19 assumptions about the processes that determine interactions (Section 3) about the levels
20 of organization at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales)
21 and the way in which we construct the resulting networks (Section 4).

22 Fundamentally, we are talking about an intersection of the type of data used to construct
23 a network and the underlying theory as to what drives the resolution and occurrence of
24 interactions among species in those data. We still lack a clear explanation of the different
25 assumptions and scale dependent processes that underpin network construction alongside
26 extensive discussions about the challenges relating to data collection and observation
27 [6,10,14–21]. Such an understanding should deliver an acceleration in capacity to more
28 effectively predict the impact of multiple stressors on biodiverse communities.

29 In their recent work, [22] showcased a 2+2 decomposition of networks around aggregated
30 versus species level resolution of nodes and around potential and realised links among the
31 nodes. Their review delivers valuable insight into the methodologies used to collect and
32 manage data among the node and link differentiation. It also delivers an overview of the
33 scale and types of questions that are associated with each category of differentiation.

34 Here we provide a complementary perspective focused on concepts, models, and theory, in
35 contrast to the data driven breakdown in [22] (*e.g.*, their Tables 1 and 2). Our approach
36 delivers a hierarchical perspective on network construction based on a gradient from
37 feasibility, capturing the concept of metawebs and Gauzen et al’s ‘potential’ webs, through
38 to realised webs as in Gauzens’ et al. In contrast to their 2 + 2 decomposition (their Fig 1),
39 our perspective showcases nested ecological scales and processes that derive from shifts in
40 the assumptions and theories embedded along this gradient. This includes classic ecological
41 ‘aggregations’ such as functional/phylogenetic groups through to species, populations
42 and individuals, unique perspective on how space and time intersect with node and link
43 resolution, refined insight into which networks are derived by induction vs. deduction and
44 a revealing of a core transition between assumptions about how links are derived based on
45 evolutionary vs. ecological theories.

46 In the following sections we provide a scene-setting review of nodes and edges (links) in
47 networks before aligning various processes that determine interactions with the different

48 network representations. Ultimately, we provide a unique perspective on the nested
49 hierarchy of processes that govern transitions from meta-webs to realised webs. We finish
50 with a refined and nuanced alignment of models/representations and key questions in
51 biodiversity science in the anthropocene.

52 **2 Setting the Scene: The Not So Basics of Nodes and** 53 **Edges**

54 Ecological networks serve multiple uses, representing an ‘object’ from which inferences
55 can be made. While many aspects of community structure can be analysed without
56 networks (*e.g.*, through trait distributions or abundance patterns) networks provide a
57 formal framework for capturing the organisation of interactions among species. The study
58 of network structure and topology has a long history in ecology, rooted in early theory on
59 energy flow [2,23], and later extended to questions of robustness, stability, and complexity
60 [*e.g.*, 21,7,24–26]. More recent work has built on this foundation to link network structure
61 to ecosystem functioning, persistence, and dynamical behaviour [*e.g.*, 27,28,29]. Networks
62 are therefore commonly treated as response variables in tests of ecological theory and
63 statistical models of the generative processes that give rise to observed structure, and are
64 widely used to compare communities across environmental gradients or through time [*e.g.*,
65 30,31]. They also provide a platform for evaluating downstream responses to perturbations,
66 including secondary extinctions and robustness to species loss [*e.g.*, 7,32–34], as well as
67 for implementing dynamical process that inference about stability, ecosystem function,
68 invasions, climate change, contaminants, and extinction cascades [*e.g.*, 35,36,37]. Against
69 this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the
70 levels of organisation at which they are defined, take many forms [17,38], each of which
71 encode a series of assumptions within a network. Here we introduce a perspective on these
72 baseline assumptions.

73 **2.1 How do we define a node?**

74 While nodes are conventionally described as representing species, in practice they may
75 correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera,
76 or families, as well as trophic species (*e.g.*, [39]; [40]), feeding guilds [*e.g.*, 41], or life-stage–
77 specific subsets of species [*e.g.*, 42]. These choices reflect differences in the level, type, and
78 consistency of resolution at which networks are constructed, rather than aggregation per se.

79 In many cases, aggregation is applied explicitly on top of more highly resolved—or unevenly
80 resolved—data to address particular questions, as in the construction of trophic species,
81 and the underlying data may remain available for alternative analyses. Nevertheless,
82 representing nodes at coarser or mixed resolutions can limit taxon-specific inference
83 (*e.g.*, whether species *a* consumes species *b*), bias estimates of degree distributions—
84 particularly generality and vulnerability—and complicate downstream analyses such as
85 extinction or invasion dynamics, where species identity and the consequences of loss may
86 be obscured [42,43]. At the same time, there are clear justifications for using aggregated
87 representations when the distribution of interactions among functional or trophic units is
88 more informative than species-level detail, for example when analysing extinction patterns
89 across feeding guilds [44]. More broadly, issues of resolution, scale, and sampling have long
90 been recognised as central to the construction and interpretation of food webs [*e.g.*, 6,20].

91 2.2 What is captured by an edge?

92 In order to break down the definitions of an edge, it is important to introduce the concept
93 of *potential* versus *realised* links: potential links reflect feasibility while realised links are
94 connected to flux of some currency (typically energy; see below for more detail). Links
95 within food webs are thus a representation of either potential links between species [45] or
96 fluxes within a system *e.g.*, energy transfer or material flow as the result of the feeding
97 links between species [2,46]. Edges can thus correspond to different ‘currencies’ [22]. There
98 are also a myriad of ways in which the links themselves can be specified. Links between
99 species can be treated as present or absent (*i.e.*, binary), may be defined as probabilities
100 [47,48] or by continuous functions which further quantify the strength of an interaction
101 [49]. How links are specified thus requires intersecting both the currency being modelled
102 and their specification. For example, feasibility is unlikely to accommodate flux, but does
103 align with binary or probability representations. Taking a food web that consists of links
104 representing feasible interactions among a collection of species will be meaningless if one
105 is interested in understanding the flow of energy through the network as the links are not
106 environmentally/energetically constrained.

107 2.3 Network representations

108 Against these definitions of nodes and edges, networks fall into two major ‘types’: metawebs,
109 traditionally defined as all the *potential* interactions for a specific species pool [20]; and
110 realised networks, which is the subset of interactions in a metaweb that are *realised* for

111 a specific community at a given time and place. The fundamental differences between
112 these two network representations are the spatial and temporal scale at which they are
113 constructed, and the associated processes that are assumed to drive patterns at these
114 scales.

115 A metaweb is, at its core, a list of *feasible* interactions between pairs of species. The
116 feasibility for a given pair is derived from the complementarity of their traits, typically
117 aligned with feeding. Feasibility can be further refined by *co-occurrence* leading to
118 the transition from a *global* to *regional metaweb*. Metawebs thus provide a means to
119 identify evolutionary plausible links, regionally plausible interactions, the set of ecologically
120 impossible, *i.e.*, forbidden, links [50], and ultimately a definition of the plausible *complete*
121 diet of a species [51].

122 In contrast, realised networks are typically localised in space and time, with links contingent
123 on species co-occurrence, environmental conditions, and mechanisms of diet choice. As a
124 result, the presence or absence of a link reflects species behaviour, such that even when
125 a realised network is represented as a binary matrix, each edge implies an underlying
126 function describing interaction strength. A realised network is therefore not a simple
127 downscaling of a metaweb based on finer spatial or temporal resolution. Rather than
128 being obtained by filtering feasible interactions through co-occurrence alone, realised webs
129 capture the processes that govern whether interactions are expressed and how energy
130 flows through a community. In this sense, the definition of an edge shifts from one of
131 feasibility to one of realised choice and consequence, centred on energy acquisition and
132 transfer. Consequently, a metaweb and a realised network constructed from the same set
133 of species may share node composition yet differ structurally, because they are constrained
134 by fundamentally different rules governing link presence. This distinction has important
135 implications for interpretation: links absent from a metaweb can be treated as truly absent
136 in terms of feasibility, whereas links absent from a realised network should instead be
137 interpreted as context-dependent absences arising from environmental, behavioural, or
138 community-level constraints.

139 3 From Nodes and Edges to Process and Constraints

140 In the previous section we discussed how the definition of nodes and edges, representing
141 different scales and processes, lead to the concept of a metaweb and a realised web. The
142 fundamental take-homes are that nodes vary in their resolution, edges vary in what kind

of process they represent and the intersection of these, defined by meta- vs. realised webs, underpins distinct lines of inquiry and constraints on the type of inference we can make with networks. Here we reveal five core constraints across evolutionary and ecological scales that further delineate the transition from meta- to realised webs, exposing processes that determine the nature of links among nodes: evolutionary compatibility, co-occurrence, abundance, diet choice, and non-trophic interactions Figure 1.

[Figure 1 about here.]

3.1 Processes that determine the feasibility of an interaction

Evolutionary compatibility and co-occurrence are the two principal processes that ‘act’ at the species pair of interest and define feasibility. The scale of inference and set of processes embodied in these two constraints typically combine to define a ‘list’ of interactions that are viable/feasible and defined strictly as present/absent. Reflecting on the previous section, nodes are typically species and rules defining edges are defined by trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight into each process.

Evolutionary compatibility

This constraint is defined by shared (co)evolutionary history between consumers and resources [52–55] which is manifested as ‘trait complementarity’ between two species [56]. In this body of theory, the consumer has the ‘correct’ set of traits that allow it to acquire and consume the resource. Interactions that are not compatible are defined as forbidden links [50]; *i.e.*, they are not physically possible and will *always* be absent within a network.

Networks do not properly arise from models based on this constraint. Instead, interacting species pairs are defined and these are represented as binary (possible vs forbidden) or probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are quantified as the confidence of a specific interaction being *possible* between two species. A network constructed on the basis of evolutionary compatibility is conceptually aligned with a ‘global metaweb’, and gives us information as to the global feasibility of links between species pairs despite the fact that they do not co-occur (see Figure 1).

(Co)occurrence

The co-occurrence of species in both time and space is a fundamental requirement for an interaction between two species to occur (at least in terms of feeding links). Although co-occurrence data alone is insufficient for building an accurate and ecologically meaningful

174 representation of *feeding links* [58], it is still a critical process that determines the possible
175 realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially
176 constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1
177 this would be the metawebs for regions one and two.

178 We reinforce that these two constraints don't deliver a network *per se*, but a list of feasible
179 species pairs. Although it is possible to build a network from the list of interactions
180 generated by these constraints, it is important to be aware that the structure of this
181 network is not constrained by any community context: just because species are able to
182 interact does not mean that they will [60,61].

183 3.2 Processes that realise networks

184 In contrast to the above, here we highlight three processes that influence the *realisation* of
185 an interaction between species and thus form the conceptual basis for realised networks.
186 As we show in Figure 1, a 'truly realised' network is the product of properties of the
187 community (**abundance** and **non-trophic interactions**) and the individual (**diet**
188 **choice**). This represents a conceptual shift from considering the feasibility for species
189 pairwise interactions to considering the edge as a representation of energy flow. Such
190 a transition requires information about how the community, the environment and the
191 individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

192 Abundance

193 Abundance as a realising process emerges from a null model for energy acquisition:
194 organisms feeding randomly will consume resources in proportion to their abundance
195 [63]. Here, abundance of different prey species influences the distribution of links in a
196 network [64] by defining a preference linked to individuals among species meeting [47,60].
197 Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can
198 define the distribution and strength of links. Of note, however, is that such abundance
199 constrained interactions are not necessarily contingent on there being any compatibility
200 between species [65–67].

201 Diet choice

202 It is well established that consumers make more active decisions than eating items in
203 proportion to their abundance [63]. Ultimately, consumer choice is underpinned by
204 an energetic cost-benefit framework centered around profitability and defined by traits

205 associated with acquisition and consumption of a resource [68,69]. Energetic constraints
206 are invoked to construct networks in a myriad of ways [*e.g.*, 43,70–72].

207 In contrast to metaweb ‘construction’ from a list of pairwise interactions, these methods
208 deliver a realised web directly and as an emergent property of node behaviour. We also
209 here make a distinction, developed below, with models like the Niche Model [40], where
210 diet choice is implicit in its probabilistic network generating function, but it is working
211 to replicate the *expected* structure of the network, and this structure does not emerge
212 from node-based rules. Note that we select diet choice as a term to capture rules linked
213 to optimal foraging [73] and metabolic theory [74]; it is a sensible ‘umbrella concept’ for
214 capturing the energetic constraint on of the distribution and strength of interactions.

215 **Non-trophic interactions**

216 We include non-trophic interactions [see 75] here not as a determinant of links, but a
217 modifier of them - they are the community context above and beyond co-occurrence and
218 abundance. Non-trophic interactions include competition for space, predator interference,
219 refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or
220 decrease mortality. These interactions specifically modify either the realisation or strength
221 of trophic interactions [29,32,76–78] and represent direct (*e.g.*, predator *a* outcompetes
222 predator *b*) and indirect (*e.g.*, mutualistic/facilitative interactions) mechanisms.

223 Some interactions, such as pollination, occupy an intermediate position in this framework,
224 as they combine trophic components (*e.g.* resource consumption) with non-trophic effects
225 that influence reproduction, recruitment, and population persistence [79–81]. They operate
226 on the realisation of a network by altering the fine-scale distribution and abundance of
227 species and relative contributions of direct and indirect effects to biomass, persistence,
228 stability and the functioning of the communities [75,82–84].

229 **4 Network construction**

230 The above five processes are central to understanding the assumptions inherent in building
231 different types of networks. Each of the processes, or combinations thereof, deliver a
232 unique set of boundary conditions on what a network represents and can be used for. Here
233 we build on the introduction of these five processes to further categorise the approaches to
234 constructing networks. In doing so also introduce more detail on a variety of methodologies
235 used to construct networks.

4.1 Why construct networks?

Networks are a representation of biodiversity. In a perfect world, we might know about all interactions. However, the empirical collection of interaction data is both costly and challenging to execute [50,85,86]. In the absence of robust empirical data, we construct models that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.*, 87,88–90], predict the feasibility of interaction among pairs of species, or directly predict network structure [see 91 for a broader discussion].

They are unique in delivering more than just estimates of species richness. As noted in the introduction, a network embodies the organising structure of biodiversity and allows numerous opportunities for ‘downstream’ analysis, including the comparison of structures, estimation of energy flux or extinction dynamics and ultimately form the structural inputs to dynamical systems models that facilitate ecological and conservation relevant inference about productivity-diversity-stability-function relationships [28] in space and time. But making such inferences requires careful attention to one or more of the processes discussed in Section 3. While these network representations may be simplifications of the truth [92], they remain critically useful tools as they allow us to move beyond simple descriptions of species richness to test hypotheses about community architecture and ecosystem functioning that would be otherwise impossible to assess.

4.2 Construction through induction

Constructing feasible or realised networks can be framed as an ‘inductive reasoning’ process where insight and generalisation arises from a set of observations and relationships. Inductive reasoning as a foundation for network construction is implemented at node and network levels. When applied at the node level, species-specific networks are created and judged by their association with expected feeding interactions. When applied at the network level, networks are judged by their structural properties.

4.2.1 Species specific networks: construction through node level induction

Constructing feasible networks and facilitating the interpolation or gap-filling of existing empirical datasets on sets of species interactions can be framed as an ‘inductive reasoning’ process where insight and generalisation arises from a set of observations and relationships about feeding. All methods in this inference space rest on a set of three assumptions: there are a set of ‘feeding rules’ that underpin interaction feasibility [93]; these rules

267 are phylogenetically conserved [54,94]; and they can be specified by matching the traits
268 between consumer and resource.

269 Evolutionary compatibility and co-occurrence constraints have been critical to the construc-
270 tion of ‘first draft’ networks for communities for which we have no interaction data [57].
271 They are also central to interpolation in data poor regions and predicting interactions for
272 ‘unobservable’ communities *e.g.*, prehistoric networks [44,95,96] or future, novel community
273 assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions
274 among species relative to their distribution by accounting for the role of the environment
275 and the role of species interactions [98–101]. There are substantial data requirements for
276 these approaches including expert knowledge, species traits and phylogenetic relationships
277 and/or interaction data on related species or communities.

278 Feeding rules are defined in multiple ways. The determination of the feeding rules can be
279 defined *a priori* based on expert knowledge opinions. Typically this is done on a ‘trait
280 matching’ basis. An example are the paleo food web models of [102] and [103] that specify
281 a series of rules for a set of traits and interactions deemed feasible if all conditions are
282 met. Alternatively the body size ratio between the consumer and resource is often used
283 [*e.g.*, 104,105], with the idea that consumers will only utilise a resource with a body size is
284 less than or equal to their own [106,107], although it is broadly acknowledged that many
285 herbivores (such as insects) violate these assumptions [108]. However, work from [109]
286 seems to suggest that adding morphological traits in addition to body size ratio improves
287 model performance.

288 Rules are also defined by correlating real world interaction data with suitable ecological
289 proxies for which data is more widely available (*e.g.*, traits) using some sort of binary
290 classifier (see [110] for an overview). These include generalised linear models [*e.g.*, 111],
291 random forest [*e.g.*, 112], trait-based k-NN [*e.g.*, 113], and Bayesian models [*e.g.*, 114,115].

292 Finally, graph embedding uses the structural features of a known network to infer the
293 position of species in an unknown network through the decomposition of the interaction
294 onto the embedding space. This decomposition relies on a combination of ecological
295 proxies (*e.g.*, traits) in conjunction with known interactions to infer the latent values of
296 species, which can then be mapped onto decomposition of a known network. See [51] for a
297 detailed review of methods and [57] for a specific example.

4.2.2 Species agnostic networks: construction through structure induction

Networks in this category are generated using rules that create non-random networks that reflect some minimal empirical knowledge of ecological networks. These can be used in a variety of ways, for example comparing the structure of realised networks to quantify relative deviations from the model structure. They are also used as a way to generate ‘realistic’ food web structures that are used for dynamical or other analyses. The determination of links between species is only implicitly linked to properties of the nodes. This means these networks are usually not species specific. Although these models are data input light, often requiring only species richness and an estimate of the number of expected links, they make clear assumptions regarding what the expectations are for network structure.

Stochastic network models use a probabilistic rule-set about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [116,117] and that network structure can be determined by distributing interactions along single dimension [the ‘niche axis’; [118]]. Typically these models parametrise some aspect of the network structure [although see 116 for a parameter-free model]. These models include the most commonly used network generator, the Niche model [40], as well as the original Cascade model [5] and the derived Nested hierarchy model [119]. Despite the fact that these networks are derived without any real world data they are still able to recover the structure of empirical networks [120]. These models often form the basis for dynamic models *e.g.*, the allometric trophic network [25,27] and bioenergetic food web models [35].

4.3 Construction through deduction

In contrast to the above approaches centred on feasibility, realised networks via methods reflecting abundance and diet choice typically rely on deductive reasoning and have a unique agenda to those above. In contrast to the inductive methods, inference about a realised network follows from a set of premises defining generative processes, often referred to as mechanisms. Typically, models that embed abundance and diet choice constraints reference theory that allows inference about the distribution and strength of interactions. Such models are ‘network topology generators’ and have a strong representation in research comparing network structures along environmental gradients and delivering inference about extinctions and energy flux. They also provide the structural backbone for dynamical

systems modelling to address questions about stability-structure-productivity-function relationships, secondary extinction dynamics, species invasion and climate change. There are two broad groups of models in this deductive category.

4.3.1 Species-specific networks

These models capture the behaviour of the nodes by explicitly taking into account the properties of the different species in the community. Which means that there is a degree of variance in which links are predicted between species unlike the more ‘static’ predictions made by inductive models. However, these networks are ‘costly’ to construct in real world settings (requiring data about the entire community, as it is the behaviour of the system that determines the behaviour of the part) and also lack the larger diet niche context afforded by metawebs.

Neutral networks are built on the assumption that foraging decisions are tied *only* to the abundance of species within the community [121,122]. Here links are solely determined by the relative abundance of the different species in the community. Although it is highly unlikely that abundance is the only determinant of interactions work by [67] showcases how these neutral processes can be used in conjunction with inductive models to construct more refined/localised networks.

There is a broader group of models that focus on determining interactions in terms of energetic constraints on diet breadth, often using the ratio of consumer-resource bodysize as a proxy for capturing the energetic constraints of feeding. Models such as those developed by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as ‘realised’ owing to the energetic constraint placed on links. A different subset of diet models [*e.g.*, 43,123] use a diet choice approach, however similar to the stochastic network models they also embed assumptions on network structure. Thus these models predict both interactions and network structure simultaneously, although they would benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility) parameterisation [36].

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/population specific measures of the effects of stress and disturbance to community level metrics of these impacts. These metrics, such as resilience and more generally stability, ecosystem function and biodiversity *per se*, are natural properties of networks. This suggests that modern conservation, risk assessment and resource management requires robust network tools to support decision making.

This is also true in the disciplines of ecology and environmental science and their focus on abundance, distribution, functions and services that biodiversity provides [4]. Major questions remain, for example, about stability-diversity-productivity relationships, the impacts of extinctions and invasions and the impacts of multiple stressors operating at multiple ecological scales. A network approach to answering these types of questions specifically allows us to evaluate how environmental gradients and anthropogenic stress map through direct and indirect effects among species in a complex community and reveal fundamental patterns and understanding of processes in the natural world.

In order to effectively use networks to aid us in answering questions about conservation/risk assessment/management and core ecological theory, we need to be mindful that we are mapping the *correct* network representation to the question of interest [22]. Notably, there are certain questions that cannot be answered using specific network representations as the scale of the question of interest is fundamentally misaligned with either the process captured by a specific network representation Section 3.1, the underlying data that is used to construct it Section 4 or both of these factors.

Here we discuss and map the different network representations shown in Figure 1 to ‘appropriate’ research questions and agendas see also 1. We also highlight some of the key methodological challenges that currently limit our conceptualisation of a ‘network’ and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Network		
Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to <i>e.g.</i> , extinction, turnover, invasion and rewilding $\checkmark$	[124]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources Δ	[125]; [126]; [127]
	Rewiring capacity/potential of species by looking at the <i>entire</i> diets of species Δ	[128]; [32]; [129]; [130]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $\times$	[60]; [131]
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ	[101]; [132]; [133]
	Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey $\checkmark$	[134]; [135]

Network		
Representation	Example Research Question	Representative Studies
Realised webs	The allocation of multiple stressors across networks Δ	[136]; [137]
	Temperature threshold to community collapse Δ	[138]; [139]
	Extinction and persistence after harvesting/invasion/extinction $\checkmark$	[140]; [141]
	Stability-diversity-productivity-function $\checkmark$	[142]; [125]
	Explicitly tying ecosystem level processes and nutrient flows to networks $\times$	[143]
	Meta communities and the idea of meta-network-communities Δ	[144]; [145]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

The interactions in global metawebs are not constrained by the realisation of specific community assemblages (or species co-occurrence). These networks provide a platform for answering questions that assume interactions *could* occur between species (feasibility) or where the potential diet breadth of species is required. Examples of appropriate research questions at this scale includes those about *hypothetical* or *novel* communities and interactions under future climate change scenarios, or the potential ‘position’ of an invasive (or re-introduced) species within a network [146]. This scale is also appropriate for a particular class of questions related to the potential (eco-evolutionary) rewiring capacity of species, and how this may help inform on the opportunities for persistence of species within new community assemblages [130]. The implicit focus on feasibility in these examples highlights that global metawebs are linked to strong proxies for targets and mechanisms of evolutionary change. This offers a network scale for modelling ecological-evolutionary dynamics and the role that both evolutionary history, natural selection, and phenotypic

402 plasticity shapes the diet breadth of species.

403 5.1.2 Regional Metawebs

404 Regional metawebs are conceptually a spatially constrained global metaweb. They make
405 explicit the co-occurrence between species. Against this backdrop, they are conceptually
406 aligned with similar questions to those posed above, with the added focus on a community
407 facing, real-world, challenge. Additionally, regional metawebs can and have been used to
408 refine and constrain species distribution model predictions, giving us more refined range
409 maps [41] or community composition under climate change scenarios, even at global scales
410 [30].

411 However we must exercise a high degree of caution when comparing structures among
412 regional metwebs, whether discrete collections or along environmental gradients. At this
413 scale, where the network embeds the assumptions of feasibility, the link distribution is
414 not meaningfully constrained by environmental factors (see Section 2.2) and it is vital to
415 disentangle structural change *per se* from the multiple processes that might determine
416 species turnover (*e.g.*, β - diversity).

417 5.1.3 Realised networks

418 Realised network are the most representative of what comes to mind when people think of
419 networks, and more specifically how we can use them to help inform on larger biogeographic
420 processes [147]. This is partially because of the popularity and legacy of generative
421 network models (like the Cascade and Niche model) which produce realised network
422 representations, and represent the ideal ‘currency’ for which to understand the constraints
423 placed on interactions/network structure by the broader community and environmental
424 context. This change in currency and context affords us the opportunity to ask questions
425 that revolve around major ecological theory - *e.g.*, community stability and resilience,
426 biodiversity dynamics, ecosystem function, structure-function relationships - and around
427 major conservation and climate change global challenges - *e.g.*, temperature change,
428 extinction dynamics, invasion impacts and reintroductions/rewilding.

429 Realised networks embody an explicit focus on the link between network structure to
430 ecosystem function. Because the structure of realised webs are isolated from turnover
431 processes they allow us to think about the propagation of change (across both time and
432 space) which allows us to ask questions about the persistence of communities and how

they respond to perturbations or stressors (at both the level of the node as well as the modification of links). For example in Section 5.1.1 we discuss how global metawebs can inform as to the rewiring *capacity* of a species, a realised network however would be better suited to capture the rewiring of networks over time as a response to changes in the environment or community.

Although the recent boom in the availability of long-term observation data is allowing us to unpack decades of insights for stability-diversity-productivity relationships for more complex communities [28] or to evaluate the impacts and efficacy of re-introductions [148], we need to be mindful that empirical interaction data is typically accumulated over time and so it compresses the transient nature of the interactions between species [10]. Thus we need to apply a degree of caution when using empirical data to construct realised networks - although there is scope to think about developing methods that will allow us to modify metawebs in such a way that their structures become more aligned with realised webs (see the next section).

5.2 Key methodological challenges

As noted above, the three types of networks help highlight longstanding methodological challenges that affect our ability to increase both precision and accuracy of inference derived from the questions we highlight above. Here we review some of these challenges and opportunities that are arising to mitigate them.

Understanding what empirical data represents: Ultimately, knowing what is right/precise/correct in an ecological network requires robust data. What does it mean when we ‘observe’ an interaction be that directly (predator actively feeding on prey) or indirectly via *e.g.*, gut or isotope analysis. A network constructed with empirical data will most likely still represent an ‘accumulation’ of feeding data making it conceptually more closely aligned with the idea of a metaweb.

The validation of network structure: Progress has been made in the development of tools and approaches of assessing how well a model recovers pairwise interactions [91,149], but we still lack a clear set of guidelines for benchmarking the ability of models to recover structure [118,150]. This makes it challenging to assess if models are capturing network structure accurately, especially if one wants to use empirical data as the ‘testing set’. Specifically, can we use a network constructed using long-term interaction observation data (*i.e.*, conceptually a metaweb) to assess the accuracy of a modelled (*I don’t know*

465 *how I feel about this word*), realised network? Additionally one needs to think about *what*
466 aspect of accurate/precise network construction is the most important - is it the structure
467 or ability to correctly predict pairwise links? In the case of attempting to construct a
468 metaweb it is important that one is accurately recovering both links that are truly present
469 and absent, however in the case of realised webs it is perhaps not that clear. Is it sufficient
470 to correctly recover structure *e.g.*, connectance or do the pairwise links also need to be
471 correct?

472 **Transitioning between metawebs and realised webs:** Currently most approaches to
473 modelling realised networks fail to explicitly account for any form of evolutionary constraint
474 (although see [109] and [68]) and we need to develop either an ensemble modelling approach
475 [151,152] or tools that will allow for the downsampling of metawebs into realised networks,
476 [*e.g.*, 153].

477 Importantly we need to think critically how the creation of either an ‘ensemble network’
478 or downsampled metaweb might change the underlying ‘currency’ of a network and thus
479 the underlying definition of the edge *e.g.* the downsampling approach developed by [153]
480 structurally constrains the network to structurally look like a realised web, but the links
481 do not represent prey choice *per se*.

482 Having a well developed framework as well as ‘fluidity’ to allow us to scale up between meta-
483 and realised food webs lays the groundwork for us to actively begin the integration of food
484 webs into the classical metacommunity-metaecosystem space [154] as well as ecosystem
485 level processes [155]. That being said developments in the community-ecosystem space
486 should be intentional about articulating a firm distinction as to what defines a ‘network’ as
487 a unit, as well as defining the logical (spatial and temporal) boundaries between networks
488 [156].

489 **Making networks more tractable in applied spaces:** There is a disconnect when
490 it comes to effectively using networks in applied spaces. This is probably at least in
491 part related to the challenges with delineating ‘boundaries’ between networks, and how
492 these relate to ‘management units’ and scales in addition to a limited interpretability
493 of network metrics, specifically how this can be applied to conservation targets/indices
494 [157]. In order to address these shortcomings we need to make an effort to more efficiently
495 map the form (structure) of a network to its function in order to identify how this can
496 effectively be integrated into policy to make it meaningful and actionable [158], all while
497 remaining mindful to ensure that we identify the appropriate analytical tools and network

498 representation [132].

499 Taken together, these challenges highlight three overarching messages. (i) Network
500 representations are inseparable from the data and assumptions used to construct them;
501 (ii) validation and benchmarking must be explicitly aligned with the intended network
502 type and question; (iii) greater conceptual clarity is needed when transitioning between
503 metawebs and realised networks, particularly in applied contexts. Explicitly articulating
504 these distinctions is essential if networks are to be used both rigorously and transparently
505 across scales.

506 6 Conclusion

- 507 1. **Network representations encode different processes:** Different network repre-
508 sentations capture distinct assumptions, data structures, and ecological processes;
509 understanding the interplay between a network’s structure and the processes it
510 encodes is critical for determining which questions it can address.
- 511 2. **Network representations are not interchangeable:** There is no universally
512 optimal network representation. The utility of a network depends entirely on how
513 well its underlying assumptions and data align with the intended scale and process
514 of analysis.
- 515 3. **Explicit assumptions are essential for evaluation:** Explicitly articulating the
516 assumptions behind network assembly facilitates a more critical evaluation of network
517 suitability and inferential power. Consequently, researchers can move beyond the
518 uncritical use and default reliance on conventional network representations.
- 519 4. **A unified framework improves application and progress:** The framework
520 presented here provides a structured basis for comparing network representations and
521 evaluating their suitability across conceptual, methodological, and applied contexts.
522 Establishing this standard is essential for preventing the misinterpretation of network
523 data and for addressing foundational questions in biodiversity science.

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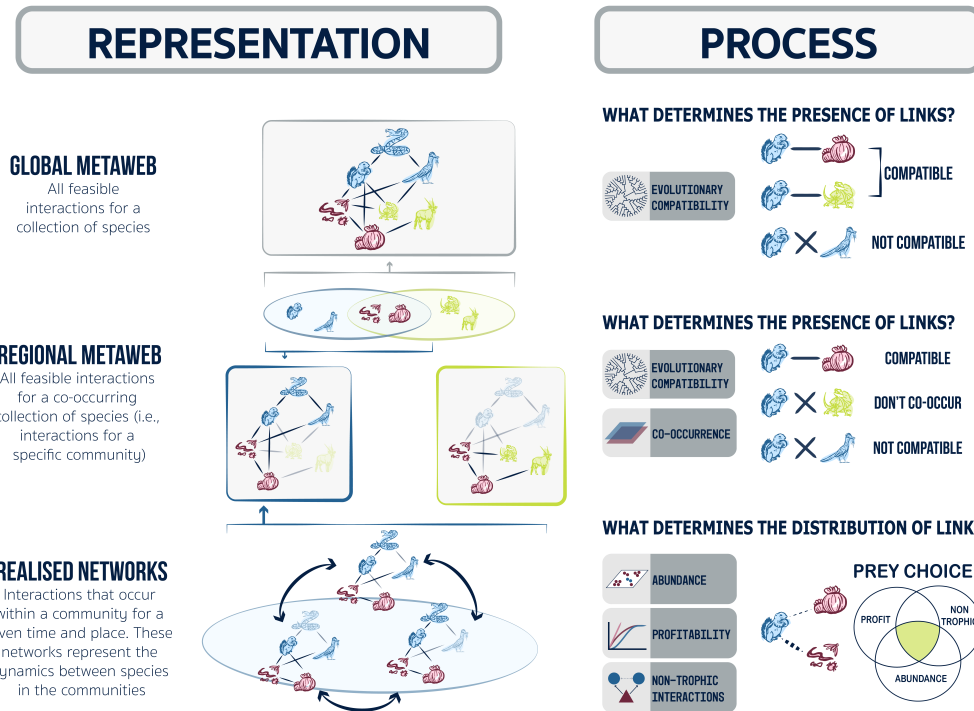


Figure 1: Aligning the various processes that determine interactions (right column) with the different network representations (left column). First, we start with a **global metaweb** this network captures all possible interactions for a collection of species in the global context. However, within the global environment different species occur in different regions (region one = yellow and region 2 = orange), and it is possible to construct two different metawebs (**regional metawebs**) for each region by taking accounting for the co-occurrence of the difference species - as shown here we have two regions with some species that are found in both regions (blue) and others endemic to either region one (yellow) or region two (orange). However, even within a region we do not expect all interactions to be realised but rather that there are multiple configurations of the regional metaweb over both space and time. The ‘state’ of the different **realised networks** is ultimately influenced not just by the co-occurrence of a species pair but rather the larger community context such as the abundance of different species, maximisation of energy gain, or indirect/higher order interactions.